AGE 310: Instrumentation and Measurements

Dr. Kadeghe Fue, Department of Agricultural Engineering, SUA

Morogoro, Tanzania

Abstract

Lecture 8 introduces the core concepts of digital electronics necessary for control: Boolean Algebra and Microcontrollers (MCUs). We explore how the continuous analog world is translated into the binary language of 0s and 1s, analyze the fundamental logic gates, and detail the architecture and programming environment of the Arduino platform, which serves as the central processing unit for many small-scale agricultural automation projects.

Lecture 8: Digital Logic and Microcontrollers

8.1 Digital Foundation: Logic and Number Systems

8.1.1 The Binary System

Computers and digital systems operate using Base-2, or the Binary System. A single digit is a **bit** (Binary Digit), which can only be 0 (LOW, 0 V) or 1 (HIGH, 5 V).

- **Byte:** 8 bits.
- **Word:** Typically 16 or 32 bits (the size of data handled by the MCU at once).

Calculation Example 1: Binary to Decimal Conversion
Convert the 8-bit binary number 10110101_2 to its decimal (Base-10) equivalent.
Solution: We use powers of 2 ($128, 64, 32, 16, 8, 4, 2, 1$):

$$(1 \times 128) + (0 \times 64) + (1 \times 32) + (1 \times 16) + (0 \times 8) + (1 \times 4) + (0 \times 2) + (1 \times 1)$$

$$128 + 0 + 32 + 16 + 0 + 4 + 0 + 1 = 181_{10}$$

8.1.2 Boolean Algebra and Logic Gates

Logic Gates are electronic circuits that implement basic Boolean functions (AND, OR, NOT). They are the building blocks of all digital processing.

1. **AND Gate ($\cdot$):** The output is HIGH only if **ALL** inputs are HIGH. (Equivalent to components connected in series).
2. **OR Gate ($+$):** The output is HIGH if **ANY** input is HIGH. (Equivalent to components connected in parallel).
3. **NOT Gate (Inverter, $\overline{A}$):** The output is the inverse of the input.
4. **XOR Gate ($\oplus$):** The output is HIGH if inputs are different (odd number of HIGH inputs).

[Image of Logic gate symbols and truth tables for AND, OR, and NOT gates]

Calculation Example 2: Digital Logic Application

A machine safety interlock circuit requires two conditions to be met for the machine to RUN (Output $Z = 1$):

1. Operator Safety Guard is CLOSED ($A = 1$).
2. Emergency Stop is NOT pressed ($B = 0$).

- Determine the Boolean expression for the required logic.
- Construct the truth table.

Solution:

- Condition 2 means the input must be $\overline{B}$, where B is the E-Stop button signal ($B = 1$ when pressed).
- The output must satisfy A AND $\overline{B}$.

$$Z = A \cdot \overline{B}$$

Truth Table for Safety Interlock

A (Guard CLOSED)	B (E-Stop Pressed)	$\overline{B}$	$Z = A \cdot \overline{B}$ (RUN)
0	0	1	0
0	1	0	0
1	0	1	**1**
1	1	0	0

8.2 Microcontrollers (MCUs)

8.2.1 MCU vs. Microprocessor (MPU)

- **Microprocessor (MPU, e.g., Intel i7):** Contains only the Central Processing Unit (CPU). Requires external memory (RAM, ROM) and peripherals (I/O). Powerful, high-speed, used in computers.
- **Microcontroller (MCU, e.g., ATmega328):** A **System-on-a-Chip** (SoC). Integrates the CPU, memory (RAM, Flash), and all necessary peripherals (ADC, GPIO, Timers) onto a single integrated circuit (IC). Low power, cheap, used for embedded control systems.

8.2.2 The Arduino Platform

The Arduino is a popular open-source prototyping platform based on Microcontrollers (like the ATmega328). It is the ideal tool for Agricultural Engineers to quickly design and test control systems.

Key Features for AGE 310:

- **GPIO (General Purpose I/O):** Pins can be configured as inputs (reading sensors/switches) or outputs (driving LEDs/relays).
- **Analog Input (ADC):** Converts $0 - 5$ V sensor signals into digital numbers.
- **PWM Output:** Used to simulate analog voltages for motor speed control or dimming lights.

8.3 Digital I/O and Control

8.3.1 Digital Input

Reading the state of a switch, limit sensor, or digital communication pin.

$$\texttt{digitalRead(pin)} \rightarrow \texttt{HIGH (1) or LOW (0)}$$

Pull-up/Pull-down Resistors: When a switch is open, the input pin is "floating" (uncertain state). A pull-up or pull-down resistor connects the pin to a known state (Vcc or GND), ensuring the reading is reliably 0 or 1.

8.3.2 Digital Output

Setting a pin to a HIGH (5 V) or LOW (0 V) state to drive actuators via relays or transistors.

$$\texttt{digitalWrite(pin, value)}$$

Current Limit: MCU pins have low current capacity (≈ 20 mA). They cannot directly power actuators like motors or solenoids; they must use a transistor or relay driver (Lecture 7).

8.4 Analog I/O and PWM

8.4.1 Analog Input (ADC)

The microcontroller's internal Analog-to-Digital Converter (ADC) converts analog voltage signals to a digital value.

```
analogRead(A0)
```

The Arduino Uno ADC has 10-bit resolution.

Calculation Example 3: Sensor Scaling

A soil pH sensor outputs a voltage range of 1 V (pH4) to 4 V (pH10). The Arduino ADC is 10-bit with $V_{\text{ref}} = 5$ V.

1. Calculate the digital reading when the pH sensor reads 2.5 V.
2. Calculate the pH value when the ADC reads 717.

Solution:

1. **Digital Reading (D):**

$$D = \frac{V_{\text{in}}}{V_{\text{ref}}} \times 1024 = \frac{2.5\ \text{V}}{5\ \text{V}} \times 1024 = 512$$

2. **Voltage from ADC reading 717 (V_{out}):**

$$V_{\text{out}} = \frac{D}{1024} \times V_{\text{ref}} = \frac{717}{1024} \times 5\ \text{V} \approx 3.5\ \text{V}$$

pH Calculation (Linear scaling required): The sensor range is 3 V (4 V $-$ 1 V) over 6 pH units ($10 - 4$). Sensitivity is $3\ \text{V}/6\ \text{pH} = 0.5\ \text{V/pH}$. Voltage above pH$4$ zero point: 3.5 V $-$ 1 V $= 2.5$ V.

$$\text{pH} = 4 + \left(\frac{2.5\ \text{V}}{0.5\ \text{V/pH}}\right) = 4 + 5 = 9.0$$

8.4.2 Pulse Width Modulation (PWM)

Since MCUs are digital, they cannot output a continuous analog voltage (e.g., 2.5 V). Instead, they use PWM to simulate it by rapidly switching a digital pin HIGH (5 V) and LOW (0 V).

Duty Cycle: The fraction of time the signal is HIGH.

$$V_{\text{avg}} = V_{\text{max}} \times \text{Duty Cycle}$$

- **Example:** 50% duty cycle means the output is HIGH half the time, resulting in an average voltage of 2.5 V.
- **Ag Use:** Controlling the speed of DC motors (Lecture 7) or dimming LED grow lights in a greenhouse.

8.5 Application: Automated Greenhouse Control

This combines all aspects of instrumentation: sensing, decision, and actuation.

- **Goal:** Maintain temperature (T) between 25°C and 30°C.
- **Sensor (Input):** Thermistor/RTD (Analog Input).
- **Actuator (Output):** Exhaust Fan (controlled by Relay, Digital Output).
- **Control Logic (Code):**
  1. Read and convert Analog T to digital value D.
  2. Scale D to temperature in $^\circ$C.
  3. **IF** T $> 30^\circ$C $\rightarrow$ digitalWrite(FAN_PIN, HIGH)
  4. **IF** T $< 25^\circ$C $\rightarrow$ digitalWrite(FAN_PIN, LOW)

8.6 Data Logging and IoT Connectivity

Data Logging: MCUs facilitate direct logging by writing time-stamped sensor data to an external SD card module (using SPI communication, Lecture 9).

Telemetry (IoT): Connecting the MCU to the internet via Wi-Fi or GSM modules allows real-time data transmission to a cloud database (e.g., Google Cloud, Thingspeak).

- **Benefit:** Farmers can monitor farm conditions remotely via smartphone (Kutoka Morogoro kwenda Mbeya).

8.7 Summary of Key Terms (Swahili/Context)

- **Binary:** Nambari za 0 na 1.
- **Logic Gate:** Lango la Mantiki (The electronic decision maker).
- **MCU:** Kidhibiti Kidogo (Microcontroller - computer on a chip).
- **GPIO:** Bandari za I/O za Kusudi la Jumla (General Purpose I/O).
- **PWM:** Urekebishaji wa Upana wa Pulsi (Pulse Width Modulation - simulates analog output).
- **Duty Cycle:** Uwiano wa Muda (Percentage of time the signal is ON).
- **Pull-up/Pull-down:** Kipinga Pull-up/Pull-down (Resistor to set a known voltage state).
- **Embedded System:** Mfumo ulioingizwa (A computer system dedicated to one control task).

Lecture 8: Review Questions

SECTION A: Basic Concepts (5 Questions)

1. Differentiate between a Microprocessor (MPU) and a Microcontroller (MCU) in terms of their key integrated components and typical applications.
2. Explain the function of a *Pull-up Resistor* when reading the state of a mechanical switch using a GPIO pin on an Arduino.
3. State the definition of *Duty Cycle* in Pulse Width Modulation (PWM) and how it relates to the average voltage output of a digital pin.
4. Define the functions of the AND gate and the OR gate. Draw the standard symbols for each.
5. Explain why a digital output pin on a microcontroller cannot directly drive a 12 V solenoid valve, even though the pin may be set to HIGH (5 V).

SECTION B: Advanced Analysis and Calculations (10 Questions)

6. **Logic Circuit Design:** Design a safety logic circuit (Boolean expression and gate diagram) for a large post-harvest grain dryer that only RUNS ($Z = 1$) if *EITHER* the door is closed (A $= 1$) *OR* the Emergency Bypass is Active (B $= 1$), *AND* the internal temperature limit is *NOT* exceeded (C $= 0$).
7. **ADC Range Scaling:** An RTD sensor outputs 2 V at 0°C and 4 V at 100°C. This is read by a 10-bit ADC with $V_{ref} = 5$ V. If the ADC reads 614, calculate the corresponding temperature in $^{\circ}$C.
8. **PWM Motor Control:** A DC motor requires 9 V average voltage to spin at 80% of its maximum speed. If the motor is driven using PWM from a 12 V power supply, calculate the required *Duty Cycle* percentage.
9. **Bit Resolution and Range:** A sensor for soil Electrical Conductivity (EC) requires a digital resolution of at least 0.05 dS/m. If the EC range is 0 dS/m to 10 dS/m, calculate the minimum *bit depth* (n) required for the ADC to meet this precision requirement.
10. **Binary Arithmetic:** Using the 8-bit binary system, convert 125_{10} to binary and perform the binary addition: $01100101_2 + 00010011_2$. Convert the result back to decimal.
11. **Pull-up Resistor Calculation:** A pushbutton switch is connected between a GPIO pin and GND. A 10 kΩ internal pull-up resistor is used (connecting the pin to 5 V). If the button is pressed, estimate the current flow through the resistor and the button, and explain why this current value is safe for the circuit.
12. **Microcontroller Timing:** An Arduino is programmed to measure soil moisture content every 15 minutes. If the total measurement and communication time takes 5 seconds, calculate the percentage of time the MCU is in an idle (SLEEP) state. Why is power management important in remote agricultural IoT systems?
13. **Data Logging Efficiency:** An IoT system logs 10 sensor readings (each 1 byte) every minute to an SD card. Calculate the total data size generated in 30 days in Kilobytes (KB) and explain why this local logging is often preferred over continuous wireless transmission in remote SUA sites.

14. **Boolean Simplification:** Simplify the following Boolean expression, which controls the VRT fertilizer application: $Z = A \cdot B + A \cdot \overline{B}$. Interpret the simplified result in the context of agricultural control.

15. **Embedded Control Decision:** A student needs to build a prototype for an automated chicken feeder. Should they use a full computer (MPU) or a microcontroller (MCU)? Justify your choice based on cost, power consumption, and task complexity.